Edge-Guided Hyperspectral Image Compression With Interactive Dual Attention

Yuanyuan Guo, Yulong Tao, Yanwen Chong, Shaoming Pan, and Miao Liu

***Abstract*— Compressing hyperspectral images (HSIs) into compact representations under the premise of ensuring high-quality reconstruction is an essential task in HSI processing. However, existing compression methods usually encode images by smoothing due to the low-frequency information occupying a prominent component in most images. Consequently, these methods fail to capture sufficient structural information, especially in low bit rates, often causing inferior reconstruction. To address this problem, we propose here an edge-guided hyperspectral compression network, called CENet, to realize high-quality reconstruction. To enhance the structural latent representation ability, the CENet model incorporates an edge extractor neural network into the compression architecture to guide compression optimization by the edge-guided loss. We propose an interactive dual attention module to selectively learn edge features, obtain the most effective edge structure, and avoid additional edge information redundancy at the same time. In the proposed CENet, the edge-guided loss and interactive dual attention module are combined to enhance the comprehensive structure of the latent representation. Concretely, interactive dual attention makes the edge extraction network focus only on moderate boundaries rather than on all edges, which enables savings on the bit rate cost and helps achieve a strong structural representation. As a result, the reconstruction quality is significantly improved. The extensive experiments on seven HSI datasets verify that our model can effectively raise the rate–distortion performance for HSIs of any type or resolution (e.g., yielding an average peak signal-to-noise ratio (PSNR) of 30.59 dB at 0.2382 bpppb, which exceeds the baseline for Chikusei by 10.99%).**

***Index Terms*— Edge-guided loss, high-quality reconstruction, hyperspectral image (HSI) compression, interactive dual attention.**

I. Introduction

COMPARED to RGB images, hyperspectral images (HSIs) describe scenes with finer spectral details using dozens or even hundreds of bands. Such a large number of spectral channels have shown great benefit to numerous imaging tasks, including classification [1], [2], segmentation [3], [4], and recognition [5] tasks. However, rich spectral information also brings heavy burdens to the limited storage capacity and the attainable transmission bandwidth, making effective compression algorithms indispensable for HSI applications.

Compared to RGB, HSI compression faces big challenges caused by the volume of details and the significant scientific property involved in their unique spatial–spectral structure. HSIs typically describe objects with fine information. However, numerous image details discourage the representation capability of the encoder from structuring image contents [6], [7], [8]. Meanwhile, HSI compression always serves for high-level image tasks (e.g., HSI classification), which seek better reconstruction quality to preserve the strategic value of HSIs.

Many algorithms have been proposed for the challenging HSI compression. In these methods, lossy image compression has been an increasingly favorable choice due to its high compression rate. With this technique, the HSI will be stored in a compact format with the demand of an appropriate rate–distortion (RD) tradeoff. Transform coding [9] (e.g., JPEG2000 [10]) and prediction coding [11] (e.g., intraprediction) are the two common approaches with the proposed various adaptations for HSIs according to different compression requirements [12], [13], [14], [15], [16], [17], [18]. Deep learning-based compression methods display significant potential [19], [20], [21], [22], [23], [24] and have been the new state-of-the-art algorithms [20]. Most of these deep learning-based methods are based on transform coding but use stacked convolutional neural networks (CNNs) rather than conventional transforms (e.g., discrete cosine transform [25] and discrete wavelet transform [13]) to transform pixels into a squeezed representation. A great majority of architectures [22], [26], [27], [28] choose an autoencoder [29] or its variety (i.e., variational autoencoder (VAE) [30]) as the main codec. However, detail degradation in lossy compression is inevitable.

The abovementioned methods mainly encode images by removing (or smoothing) several visually imperceptible components (e.g., edge information), which may contain strong structural information of images. Actually, removing such strong structural information will heavily affect the compression performance. All of these methods significantly degrade the quality of the reconstructed images as the compression ratio increases, resulting in different compression artifacts. Traditional methods display algorithm-specific artifacts around the edges and important details, such as ringing and

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blotches. Meanwhile, deep learning-based methods reveal various issues caused by the different distortion metrics used in network training, such as the blurry reconstructions yielded with the mean squared error (mse), poor text reconstructions caused by the multiscale structural similarity index metric (MS-SSIM) [31], and checkerboard artifacts when using learned perceptual image patch similarity [32] and encoding operators (e.g., downsampling).

To solve this issue, one conventional method is transmitting additional side information to guide optimization instead of a blind RD tradeoff. Many attempts have been made by utilizing inherent features within images, such as synthesis, structure, and texture, which have been illustrated effectively in preventing compression artifacts [33], [34]. Among these methods, edge-aware approaches have presented encouraging results due to the remarkable ability of describing the image structure [35] and have combined with many downstream tasks to boost performance [36], [37], [38]. Besides, a sharply depicted edge can highly advance the spatial structure characterization [33], thereby helping understand the entire content of a scene, even at a very high compression ratio. In the context of learning-based compression methods, Mishra et al. [39] added an edge-aware loss function in the RD optimization to reduce blurring. However, this is RGB oriented compression method, and the preprocessing and postprocessing modules are designed using CNNs around a standard JPEG. All these methods are based on handcrafted codecs (e.g., JPEG2000), which ignore the joint optimization of each component, thereby causing inferior performance. Compared to RGB images, introducing all edges will unavoidably increase the bit rates during compression for HSIs because they have finer details. Therefore, we need to comprehend if all edge information is useful for compression, if not, which should be preserved. Indeed, not all obtained edges can advance the RD performance. Actually, some edges have little contribution for reducing distortion but increasing bit rates. However, insufficient edges may also fail to supply a strong structural representation; thus, a high-quality reconstruction cannot be obtained. Consequently, we need adaptively preserve edges that can promote compression performance.

We propose here a novel HSI compression network (CompNet) embedded with an edge extraction network, called CENet, which can be jointly optimized. The CENet architecture contains two main subnetworks: a CompNet implemented with a VAE and an edge extraction network (EdgeNet) implemented with a revised VGGNet. We append two extra modules, namely, the interactive dual attention and edge-guided loss function, to improve the structural ability of the compressed latent representation and, at the same time, reduce the edge bit rates as much as possible. Interactive dual attention is used to help the EdgeNet adaptively capture sufficient context concerning powerful continuous feature relations within minimal representation. Specifically, the edge extraction network delivers the object boundaries to the CompNet to assist it to understand the content of a scene. The CompNet then feeds back to the edge extraction network to focus on some edges that are helpful for distinguishing different locations for bit rate allocation as driven by the edge-guided loss, where edge-guided loss is used as guidance in the RD optimization. This function jointly optimizes the compression and edge extraction networks by minimizing the difference between the predicted edges obtained from the original HSI and the reconstruction. By integrating the interactive dual attention and edge-guided loss modules in CENet, the edge extraction network enables the CompNet to be optimized with carefully selected edge information, which is different from the existing edge-aware codecs that usually use fixed edge map for compression guidance.

CENet is an elegant network that can be used to improve the compression performance and can be easily deployed to any existing learning-based compression benchmarks. The main contributions of this article are given as follows.

1) Considering the benefit of edges for image structure description, we propose a novel compression architecture embedded with an edge extraction neural network (i.e., CENet) for HSI compression to achieve high-quality reconstruction. It has the ability to distinguish what is useful edge for compression performance, which is not possessed by traditional edge-aware codecs.
2) Interactive dual attention is proposed to selectively transfer the structural details obtained from the original HSI to the compressed HSI to capture rich context that can provide strong continuous feature relations for RD optimization, thereby avoiding extraneous bit rates.
3) The proposed edge-guided loss drives the CompNet to focus on some edges that are helpful for distinguishing different locations for bit rate allocation in a soft manner to implicitly guide the CompNet optimization.
4) The proposed CENet is validated in seven datasets with various scenes (i.e., agriculture, daily scenes, and buildings). The results show that CENet can achieve a state-of-the-art compression performance.

The remainder of this article is organized as follows. Section II briefly reviews the related work. Section III introduces CENet. Section IV discusses the compression performance of CENet. Finally, Section V draws the conclusions.

II. Related Work

Edges have shown superior advantage in image processing [3], [35], [40] due to their powerful capability in object boundary localization. This section first briefly introduces the typical edge-based works for image compression. Next, the associated edge-based HSI compression approaches are provided.

A. Edge-Guided Image Compression for High-Quality Reconstruction

Detail degradation in lossy compression seriously affects the image reconstruction quality as the compression ratio increases. Extensive works have focused on additional information for handling this problem. Among existing side information, edges have led to promising results in image compression due to their powerful ability of structure capturing. Mainberger et al. [35] illustrated that the edge locations combined with the color values or adjacent gray were sufficient

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for reconstructing high-quality cartoon-like images. Schilling and Cosman [33] showed that users could have a better understanding of the image content with some enhanced edges, which was beneficial to high-quality reconstruction, even at high compression ratios. The main insight of these methods is to convey additional structural information to the decoder to guide the RD optimization. Intuitively, edge detection is an important part of edge-based compression methods. It aims to determine an appropriate principle to distinguish object boundaries and salient edges from images. Traditional edge detection methods (e.g., canny [41]) are mostly based on some local low-level cues (e.g., gradient, color, and brightness) to define edges and boundaries. However, they may introduce false edges due to their handcrafted parameter settings. The excellent capability of CNNs in high-level feature learning and semantical representation for images recently facilitates a new trend for edge extraction (e.g., DeepEdge [42], HED [43], and RCF [44]). In this study, we also build an edge extraction network with CNNs, in which downsampling is removed to avoid detail degradation.

B. HSI Compression

As a special image case, HSIs own numerous details or features, thereby leading to higher compression requirements to keep sufficient structure information in the compressed image. Some efforts have been made to mitigate this problem. For instance, Jain and Adjeroh [45] proposed an edge-based prediction for lossless hyperspectral compression, in which edges were used as a kind of context information to separate different homogeneous regions for prediction. The pixel of interest (POI)-preserving-based lossy HSI compression described in [46] enhanced the quality of reconstructed HSIs by keeping the main high-frequency information (edges). CNN-based compression approaches have also seen great interest for HSIs. Deng et al. [47] proposed a novel HSI compression scheme with generative adversarial networks (GANs). Although GANs have been efficient in preventing compression artifacts, they may not adapt to structure preservation. Theoretically, the generator can produce images that are very different from the input. Guo et al. [24] proposed a learned hyperspectral compression with Student's T hyperprior to adapt to the non-Gaussian characteristic implied in the compressed HSIs. However, the reconstruction at low bit rates is inferior. Therefore, in view of the great potential of edges and CNNs, we designed the CENet to realize high-quality reconstruction for HSI compression.

III. Edge-Guided CompNet With Interactive Dual Attention

In this section, the proposed CENet and its two novel designs will be discussed in detail. We will first start with the overview of CENet. Two novel designs, namely, interactive dual attention and edge-guided loss, are then presented.

A. Overview

The proposed CENet tackles HSI compression from a novel perspective of edge-guided optimization. It achieves high-quality reconstruction with two new designs, including interactive dual attention and edge-guided loss. As shown in Fig. 1, CENet involves two subnetworks, the CompNet and the EdgeNet, in the main architecture edge-guided CompNet, which learns to compress HSI and extracts edges from the original HSI and the reconstruction, respectively. In addition, CENet includes an interactive dual attention module in the middle of CompNet and EdgeNet to help the EdgeNet adaptively capture sufficient context concerning powerful continuous feature relations within minimal representation, which corresponds to edge adjustment, the orange arrow with "edge adjustment" in Fig. 1. After all, not all edge information is helpful for compression performance since some edges have little contribution for reducing distortion but only increasing bit rates. Moreover, the CENet is trained under the guidance of the edge-guided loss, which strives to refine such boundaries that benefit for the RD performance. With the help of the selected edges, the CompNet can distinguish different locations for bit rate allocation, thereby realizing distortion adjustment, which is denoted by the purple arrow with "distortion adjustment" in Fig. 1. The coordination of the interactive dual attention and the edge-guided loss can be used to obtain an optimal RD performance with high-quality reconstruction in our proposed unified CENet architecture. Different from existing works, the proposed CENet treats different edges distinctively and can dynamically use edges as guidance. We will discuss the CENet details in the following.

B. Edge-Guided CompNet

Fig. 1 presents the edge-guided network architecture with two subnetworks. The first is CompNet (left side in Fig. 1) served for compression task. The second is EdgeNet, which includes two same entities separately lying on the top and bottom of Fig. 1 and is used as guidance to provide rich structural information of the compressed HSI for high-quality reconstruction. This section presents the details of these two subnetworks.

1) CompNet: Numerous deep learning-based image compression methods are available, and the most popular approach among which is the variational image compression with a scale hyperprior proposed in [28]. However, almost all these methods are explored for RGBs. In this study, we revised the input and the corresponding convolution operations to make it adaptable to the HSIs. The CompNet has two autoencoders. The first autoencoder is the main autoencoder used as encoder and decoder blocks (E and D, respectively, in Fig. 1) to learn a discrete latent representation of HSIs. The second autoencoder is a hyperprior network involved with hyperencoder (HE) and hyperdecoder (HD) blocks in Fig. 1) used for learning a probabilistic model (also referred to as entropy model) over discrete latents on which entropy coding relied upon. In this study, the hyperprior network is used to generate the mean μ and the scale σ^2 parameters for the entropy model, which is modeled as a conditional Gaussian. Between the encoder and decoder blocks (or hyperencoder and hyperdecoder blocks) are quantization (Q in Fig. 1) and arithmetic encoder (AE) and arithmetic decoder (AD) in Fig. 1.

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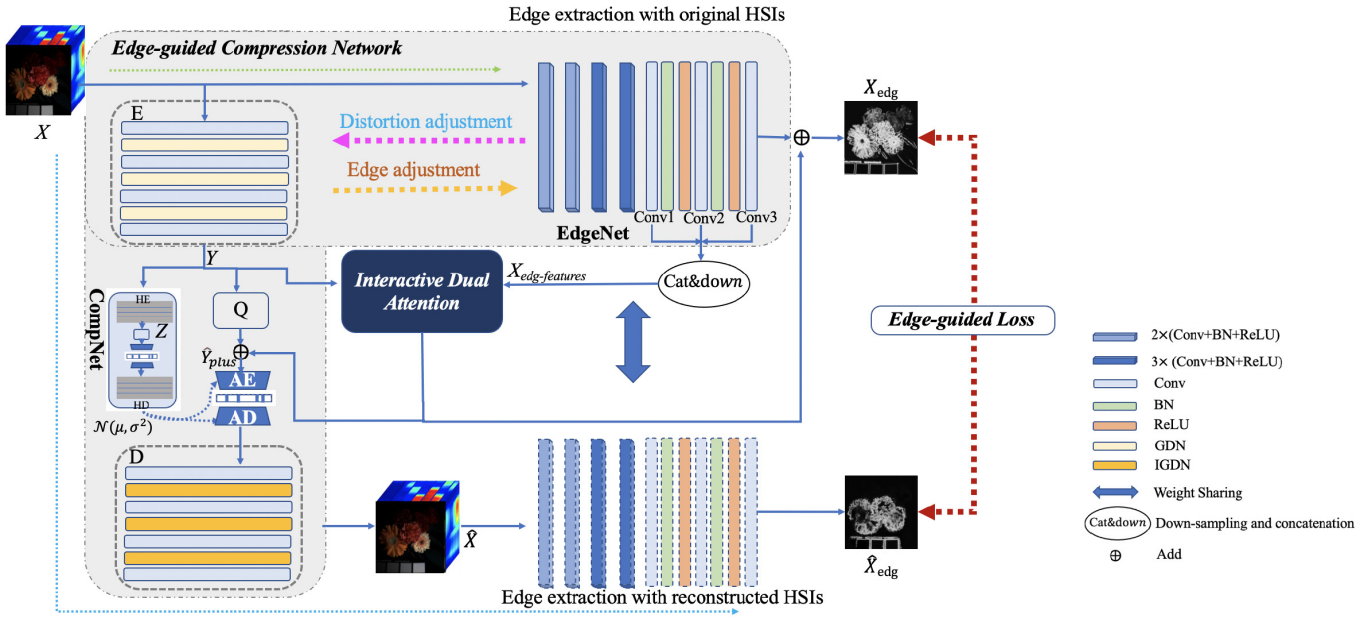


Fig. 1. Architecture of the proposed CENet.

An HSI is usually denoted as $X \in \mathbb{R}^{H\times W\times C}$, where H and W indicate the height and the width of the spatial resolution, respectively, and C represents the number of spectral bands. Encoder blocks transform X into a latent representation Y and then produce latents $\hat{Y}$ and hyperlatents $\hat{Z}$ (obtained by HE) after quantization, which is next encoded into bitstream with AE and decoded via AD. The HSI can also be recovered as $\hat{X}$ from the decoded bitstream after the decoder blocks by the receiver. Following [48], the whole CompNet optimization can be formulated by an RD function:

$$\mathcal{L}_{\text{CompNet}} = R + \lambda \cdot D = \underbrace{\mathbb{E}_{X\sim p_X}\left[-\log_2 p_{\hat{Y}}(Q(g_a(X)))\right]}_{\text{rate(latents)}} + \underbrace{\mathbb{E}_{X\sim p_X}\left[-\log_2 p_{\hat{z}}(\hat{z})\right]}_{\text{rate(hyper-latents)}} + \lambda \cdot \underbrace{\mathbb{E}_{X\sim p_X}\left[d\left(X,\hat{X}\right)\right]}_{\text{distortion}} \tag{1}$$

where λ is a balance parameter used for determining the expected RD (RD loss) tradeoff, Q represents quantization, which is implemented by rounding to the nearest integer, $Y = g_a(X)$ is obtained by the encoder, $Z = h_a(Y)$ is obtained by the HE, p_X is the unknown HSI distribution, $p_{\hat{Y}}$ denotes the learned entropy model modeled as a conditional Gaussian convolved with a standard uniform following the work of [28], $(\mu, \sigma^2) = h_s(\hat{Z})$ is the output of the HD, and $p(\hat{Z})$ is often modeled as a nonparametric, fully factorized density due to its unknown prior.

The first two terms are encoding bit rates corresponding to cross entropies when encoding the quantized latents $\hat{Y} = Q(Y)$ and hyperlatents $\hat{Z} = Q(Z)$. Ideally, these bit rates can achieve the minimum when p_X and $p_{\hat{Y}}$ are identical. The third term is distortion, where $\hat{X} = g_s(\hat{Y})$ is the reconstructed HSI and $d(\cdot)$ denotes the distortion metric (e.g., mse). During training, we replace the quantization with additive uniform noise as in [28] to optimize the CompNet with stochastic gradient descent because rounding to the nearest integer produces zero gradients almost everywhere.

2) EdgeNet: Although deep learning-based compression methods are promising, the degraded reconstruction in a low bit rate cannot be ignored. The salient edges and the object boundaries from HSIs are helpful for capturing rich data structure, thereby being used as guidance to distinguish different locations for bit rate allocation. Hence, we embedded EdgeNet into CompNet associated with the interactive dual attention as driven by the edge-guided loss to adaptively preserve edge features that can improve the quality of the reconstruction and avoid introducing too much additional bit rates.

Generally, the common choice is to generate edges with existing edge detection algorithms (e.g., canny [41]). However, these methods cannot dynamically adjust edge information during compression training since they are unable to be jointly optimized with compression task, thereby failing to distinguish what is useful edge for compression performance. Moreover, traditional edge extraction methods must manually set parameters, which may encounter great difficulties when dealing with complex HSI scenes. Therefore, we proposed here a CNN-based network for edge extraction. As shown in Fig. 1, there are two entities of the EdgeNet which share the same structure. The top entity is used to extract edge features from the original images, while the bottom is used to produce edge features from the reconstruction.

In fact, the EdgeNet is a revised VGGNet appended with three convolution layers. Table I presents the details of each layer of EdgeNet, where Conv denotes convolution, followed by the kernel size and the number of filter channels (e.g., the convolutions in the first layer of block1 use 3×3 kernels with 64 filter channels); BN stands for the batch normalization; and ReLU is the activation function. We use the first four layers of VGG16 but remove the max-pooling and downsampling operators due to their detail degradation. For the EdgeNet

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TABLE I
DETAILS OF THE INDIVIDUAL LAYERS

Input HSI			
Block 1	Conv: 3×3 −64	BN	ReLU
	Conv: 3×3 −64	BN	ReLU
Block 2	Conv: 3×3 −128	BN	ReLU
	Conv: 3×3 −128	BN	ReLU
Block 3	Conv: 3×3 −256	BN	ReLU
	Conv: 3×3 −256	BN	ReLU
	Conv: 3×3 −256	BN	ReLU
Block 4	Conv: 3×3 −512	BN	ReLU
	Conv: 3×3 −512	BN	ReLU
	Conv: 3×3 −512	BN	ReLU
Layer 1	Conv: 3×3 −128	BN	ReLU
Layer 2	Conv: 3×3 −16	BN	ReLU
Layer 3	Conv: 3×3 −1	BN	ReLU

training, we generate edge labels by canny as supervision and use the edge loss, called the annotator-robust loss (ARL), proposed in [44]). Note that the generation of edge labels does not participate in the model learning. The edge labels generated by canny are just used to guide the EdgeNet toward edge learning rather than other features. In this case, the edge label seems like a seed in the EdgeNet learning where the quality of the seed is not important, just like some methods need to give a random initialization value, and the value of the random initialization does not affect the performance of the model.

C. Interactive Dual Attention

The objects in HSIs are generally characterized with finer details than RGBs, making it difficult for encoders to describe them in a compact latent representation. To address this problem, the CompNet must have robust structural information of image contents. Indeed, not all obtained edges can advance the RD performance, which will be illustrated in the ablation experiments in Section IV. Actually, some edges have little contribution for reducing distortion but increasing bit rates. This section presents the details of how to obtain the useful edge information for compression performance with the proposed interactive dual attention. Since different latents possess different distortions of high-frequency details during encoding, we employ interactive dual attention to construct the interaction between CompNet and EdgeNet. It enables EdgeNet to focus on parts of edge information of the original HSIs that is helpful in the RD optimization for CompNet. As shown in Fig. 1, there are two inputs, the edge features and the compressed latent representation, for the interactive dual attention. For edge features $X_{\text{edg-features}}$, we make a concatenation of the last three convolution features (which are denoted as conv1, conv2, and conv3 in Fig. 1) from the EdgeNet model to enhance the detail of boundaries obtained from the original HSIs. While the latent representation Y is produced by the main encoder of the CompNet model. Given the latent representation Y and the associated edge features $X_{\text{edg-features}}$, the EdgeNet can generate rich context concerning powerful continuous feature relations for the CompNet by

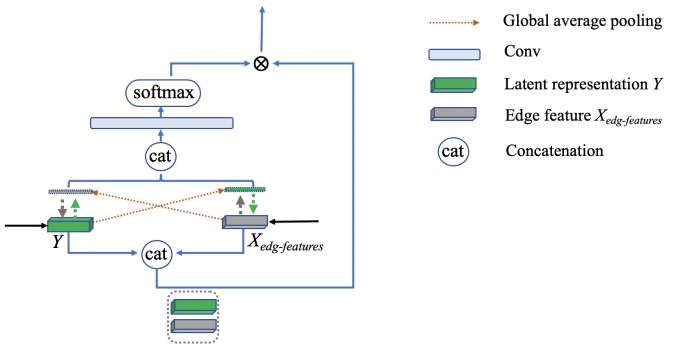


Fig. 2. Architecture of the interactive dual attention module.

interactive dual attention with the guidance of compression performance.

Fig. 2 shows the architecture of the proposed interactive dual attention module. We first implement a global average pooling for the latent representation Y and the associated edge features $X_{\text{edg-features}}$ along the spatial dimension and then build the interaction with the following functions:

$$X_{\text{edg-features}} = \langle \text{con}\upsilon\ 1, \text{con}\upsilon\ 2, \text{con}\upsilon 3 \rangle \tag{2}$$

$$I_{X_\text{edg}\to Y} = \tanh\left(Y + \Delta\left(X_{\text{edg-features}}\right)\right) \tag{3}$$

$$I_{Y\to X_\text{edg}} = \tanh\left(X_{\text{edg-features}} + \Delta(Y)\right) \tag{4}$$

$$m = = \text{softmax}\left(w^T \langle I_{Y\to X_\text{edg}}, I_{X_\text{edg}\to Y}\rangle + b\right) \tag{5}$$

where the weight vector m connects Y and $X_{\text{edg-features}}$ together, $\langle\cdot\rangle$ indicates concatenation, Δ represents global average pooling, and w and b are learned by a CNN with 1×1 kernels. Consequently, the context vector using interaction m can be obtained by

$$M_{\text{inter}} = \langle Y, X_{\text{edg-features}}\rangle \otimes m. \tag{6}$$

In (3) and (4), the structural information of latents and the associated edge features of the original HSIs are calculated as mutually dependent and interact with each other heavily. The latent representation Y is subject to the global edge vector $\Delta(X_{\text{edg-features}})$, and at the same time, the edge information is conditioned on the global latent representation $\Delta(Y)$. Intuitively, the proposed interactive dual attention module strengthens the structural ability of the latent representation with an aligned edge structure obtained from the original HSIs driven by compression performance (i.e., $\hat{Y}_{\text{plus}} = \hat{Y} \oplus M_{\text{inter}}$ in Fig. 1).

D. Edge-Guided Loss

In HSI lossy compression, although removing amount of information can effectively improve the compression ratio, the degraded details will weaken the structural representation ability of objects during decoder, thereby causing a blurry reconstruction. Moreover, the distortion metric (mse) used in network training will further aggravate this issue [22], [49] since the real distribution of HSIs is complex instead of a unimodal Gaussian. We introduce an edge-guided loss in the RD optimization to alleviate this problem. It drives the EdgeNet to focus on localizing such object contours that

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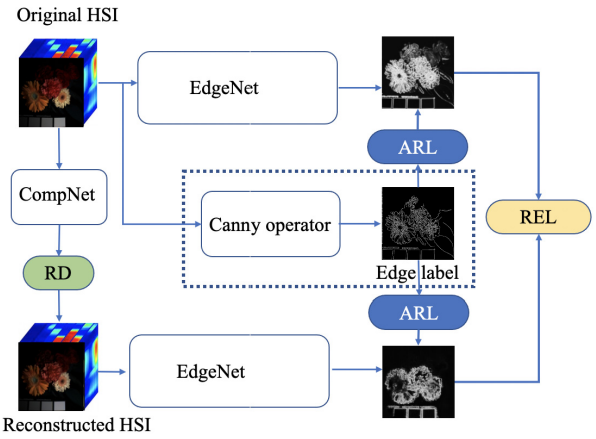


Fig. 3. Components of edge-guided loss.

benefit for compression performance from the original HSI. There are four components in the proposed edge-guided loss, an RD loss derived from CompNet, two standard ARL, and a residual edge loss (REL), which is shown in Fig. 3). The compression loss $\mathcal{L}_{\text{CompNet}}$ is formulated as (1). ARLs have two components $\mathcal{L}_{\text{ARL_}X_{\text{edg}}}$ and $\mathcal{L}_{\text{ARL_}\hat{X}_{\text{edg}}}$ to learn a consistent global structure from pixel level.

Take $\mathcal{L}_{\text{ARL_}X_{\text{edg}}}$ as an example

$$\mathcal{L}_{\text{ARL_}X_{\text{edg}}} = \sum_{i=1}^{|\Gamma|} \left(l_{\text{ARL}}\left(x_{\text{edg}_i}, \Phi\right)\right) \tag{7}$$

where

$$l_{\text{ARL}}\left(x_{\text{edg}_{i'}}\Phi\right) = \begin{cases} \xi \cdot \log\left(1 - P\left(x_{\text{edg}_{i'}}\Phi\right)\right), & \text{if } g_i = 0 \\ 0, & \text{if } g_i \in (0, 0.5] \\ \varrho \cdot \log\left(P\left(x_{\text{edg}_{i'}}\Phi\right)\right), & \text{otherwise} \end{cases} \tag{8}$$

in which $\xi = \eta \cdot (|G^+|/|G^+| + |G^-|)$, $\varrho = (|G^-|/|G^+| + |G^-|)$, G^+ and G^- indicate the positive and negative components of the edge ground truth, respectively, η is a hyperparameter used to balance the number of positive and negative components, x_{edg_i} and g_i correspond to the predicted edge value of the original HSI (features obtained from CNNs) and the edge label probability at the ith pixel, respectively, Φ represents all the learnable parameter sets, $P(\cdot)$ denotes the sigmoid function, and $|\Gamma|$ denotes the number of pixels in image X.

In addition, a residual edge REL of the reconstructed image with respect to the original helps EdgeNet learn the global structure from the semantic level, which is represented as an mse as

$$\mathcal{L}_{\text{REL}} = \mathbb{E}\left\| X_{\text{edg}} - \widehat{X}_{\text{edg}} \right\|_2^2 \tag{9}$$

where X_{edg} denotes the edges predicted by the original HSI, $\widehat{X}_{\text{edg}}$ denotes the edges predicted by the reconstruction.

Consequently, the edge-guided loss can be defined as

$$\begin{aligned} \mathcal{L}_{\text{edge-guided loss}} &= \mathcal{L}_{\text{CompNet}} + \alpha \cdot \mathcal{L}_{\text{ARL_}X_{\text{edg}}} \\ &\quad + \beta \cdot \mathcal{L}_{\text{ARL_}\hat{X}_{\text{edg}}} + \gamma \cdot \mathcal{L}_{\text{REL}} \end{aligned} \tag{10}$$

where α, β, and γ are three hyperparameters used to balance the edge guidance level for the CompNet. We set $\alpha = 1$ and $\beta = \gamma = 0.0001$ in our experiments.

IV. Experimental Results and Analysis

We conducted our CENet on seven datasets comprising five public remote sensing HSI datasets [i.e., Chikusei, Botswana, Washington DC, Pavia (Pavia Centre and Pavia University), and Houston (DFC (2013) and DFC (2018)], a public natural HSI (i.e., CAVE [50]), and a large-scale real agricultural dataset (i.e., Bil captured in Nanning, Guangxi, China[1]) to comprehensively illustrate its promising compression performance. The commonly used handcrafted HSI compression methods (i.e., KLT+JPEG2000 [14], 3D-Speck [16], and EMPR [18]) and several deep learning-based methods were also compared with CENet.

A. Dataset

In our experiments, we used seven datasets involving nine images (i.e., Chikusei, Botswana, Washington DC, Pavia (Pavia Centre and Pavia University), Houston (DFC2013 and DFC2018), CAVE, and Bil).

1) Chikusei: The Chikusei image taken by the Headwall Hyperspec-VNIR-C imaging sensor shows the agricultural and urban areas in Chikusei and covers a wavelength of 363–1018 nm. The spatial size of this image is 2517 × 2335 with 128 spectral bands. The ground sampling distance is 2.5 m. Fig. 4(a) shows its false color.

2) Washington DC: The Washington DC comprises airborne hyperspectral data over the Washington DC Mall covering a wavelength of 400–2400 nm. The spatial size of this image is 1280 × 307 with 191 spectral bands. Fig. 4(b) presents its false color.

3) Botswana: Botswana acquired by the NASA EO-1 satellite shows scenes over Okavango Delta, Botswana, and covers 400–2500-nm wavelength. The spatial size of this image is 1476 × 256 with 145 spectral bands. The spatial and spectral resolutions are 30 and 10 nm, respectively. Fig. 4(c) presents its false color.

4) Houston: The Houston dataset captured by the Compact Airborne Spectrographic Imager in different times consists of two available images, namely, DFC2013 and DFC2018. DFC2013 has a spatial size of 349 × 1905 with 144 spectral bands. DFC2018 has a spatial size of 1202 × 4172 with 50 spectral bands. Fig. 4(d) and (e) presents their false color.

5) Pavia: The Pavia Centre and Pavia University are both captured by the Reflective Optics System Imaging Spectrometer sensor. Pavia Centre has a spatial size of 1096 × 715 with 102 spectral bands, while Pavia University has a spatial size of 610 × 340 with 103 spectral bands. Fig. 4(f) and (g) shows the false color for both.

6) CAVE: CAVE is taken by a generalized assorted pixel (GAP) camera range from 400 to 700 nm. This dataset has 32 scenes over a wide variety of objects and materials (e.g.,

[1]These data have not been made public at present. Please connect with us if you need this dataset in your research.

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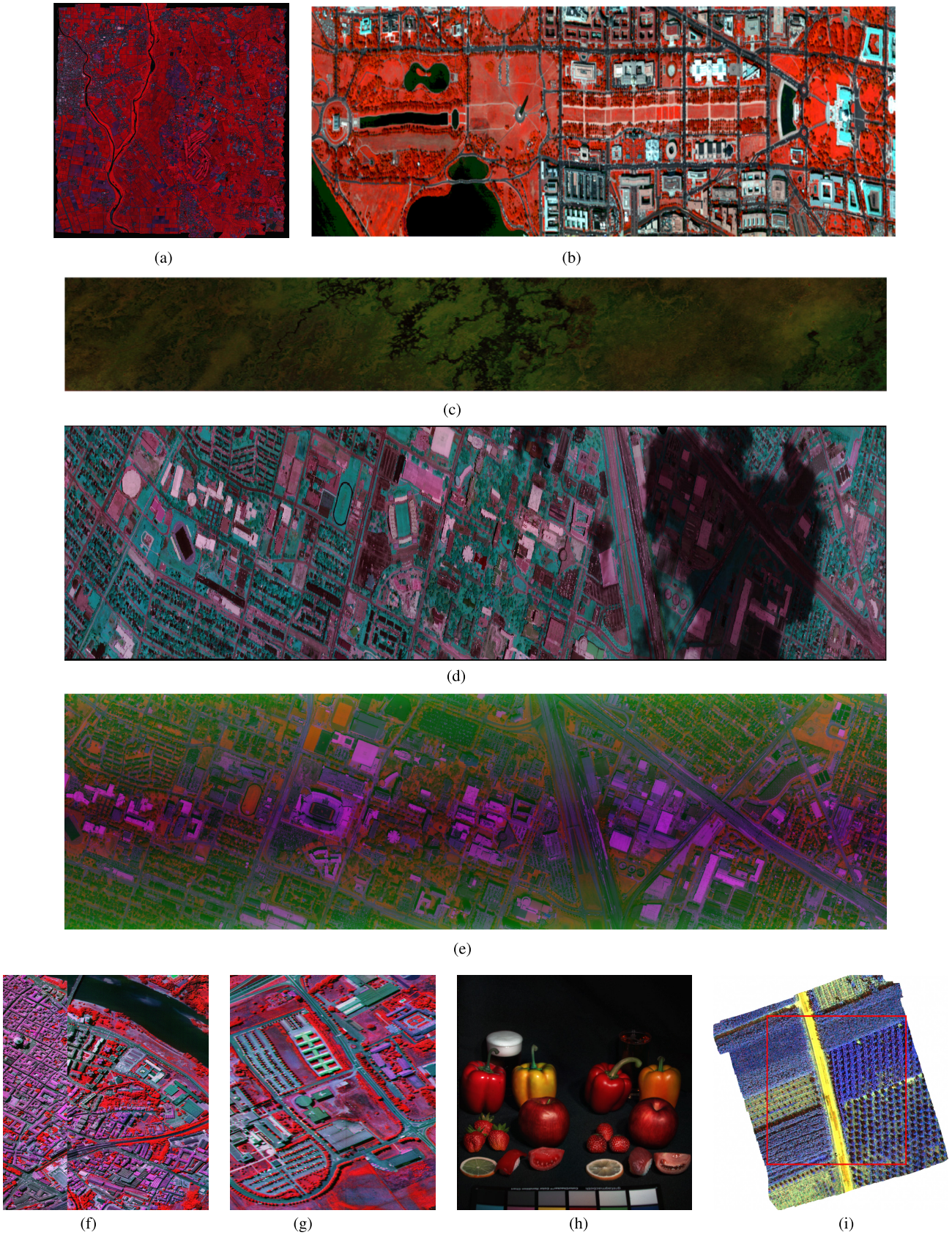


Fig. 4. (a) Chikusei data. The false color is made with bands 100, 10, and 50. (b) Washington DC data. The false color is made with bands 60, 27, and 17. (c) Botswana data. The false color is made with bands 17, 50, and 80. (d) DFC 2013 data. The false color is made with bands 50, 140, and 80. (e) DFC 2018 data. The false color is made with bands 30, 49, and 15. (f) Pavia Centre data. The false color is made with bands 100, 10, and 50. (g) Pavia University data. The false color is made with bands 100, 10, and 50. (h) CAVE data. The false color is made with bands 15, 5, and 30. (i) Bil data. The false color is made with bands 2, 22, and 86.

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textiles, skin, hair, and real and fake fruits), each having a spatial size of 512 × 512 with 31 spectral bands. The spectral resolution is 10 nm. Fig.4(h) shows a representative thumbnail.

7) Bil: Bil is a large-scale real agricultural dataset captured in Nanning, Guangxi, China. Agricultural cultivation is dominated by tropical crops, including macadamia, cassava, orange, and mango, among many others. The image spatial size is 8277 × 8041 with 126 spectral bands. The spectral resolution is 8 nm. Fig. 4(i) shows its false color.

B. Evaluation Metrics

Bit per pixel per band (bpppb) is used to evaluate the compression ratio of HSIs. In terms of distortion, we use three common metrics, namely, peak signal-to-noise ratio (PSNR), MS-SSIM [31], and spectral angle mapping (SAM) [51] to comprehensively evaluate the reconstruction quality.

1) Peak Signal-to-Noise Ratio : PSNR compares the reconstructed image to the original image from the pixel level, where a large value denotes high reconstruction fidelity. The average PSNR formulation is presented as follows:

$$\mathrm{PSNR}\left(X, \hat{X}\right) = \frac{1}{C}\sum_{i=1}^{C} 10\log_{10}\left(\frac{\max^2\left(X^i\right)}{\mathrm{mse}_i}\right) \tag{11}$$

where $\mathrm{mse}(X, \widehat{X}) = (1/H \times W \times C)\|X - \widehat{X}\|_F^2$, $\max^2(\cdot)$ denotes the square of the maximum pixel in the ith band, and C is the band number of the HSIs.

2) Multiscale Structural Similarity Index Metric: MS-SSIM is related to the difference between the original and reconstructed images in terms of the image details incorporated at different resolutions. In addition, the range of the MS-SSIM value is from 0 (worst) to 1 (best). The details are defined in [31]. Following [28], we converted the MS-SSIM value d with $-10\log_{10}(1-d)$ to make the comparison clearer.

3) Spectral Angle Mapping : SAM is used to evaluate the quality from the spectral perspective, in which a small value reflects a high spectral fidelity. The formulation is presented as follows:

$$\mathrm{SAM}\left(X, \hat{X}\right) = \frac{1}{H\times W} \times \sum_{j=1}^{H\times W} \cos^{-1}\left(\frac{x_j \odot \hat{x}_j}{\|x_j\|\|\hat{x}_j\|}\right) \tag{12}$$

where $\odot$ represents the inner product, $\|\cdot\|$ denotes the L_2 norm, and x_j and $\hat{x}_j$ denote the jth pixel of the original and reconstructed images, respectively.

C. Setup

1) Dataset Processing: We divide the seven datasets into three groups: 1) CAVE with all 31 spectral bands; 2) Bil with all 126 spectral bands; and 3) HSImix with 50 spectral bands since this is the minimum spectral band number of remote sensing HSI datasets. Training, validation, and test set partitioning are divided into three steps. First, we randomly cropped these images into nonoverlapping 512 × 512 patches. Next, we randomly selected 10% of these patches as a test set. Finally, we divided the remaining patches into training and validation sets at a 9:1 ratio. For the CAVE dataset, we randomly selected four images as the test set, randomly cropped the other 28 images into overlapping 256 × 256 patches, and then partitioned them as training and validation sets. Moreover, we first cropped a 4000 × 5000 regular rectangle from the Bil dataset [area plotted with a red rectangle in Fig. 4(i)] and then cropped it into 72 nonoverlapping patches with a spatial size of 512 × 512, where 64 patches consisted of the training and validation sets and eight patches made up of the test set. In addition, five datasets involving seven images [i.e., Chikusei, Botswana, Washington DC, Pavia (Pavia Centre and Pavia University), and Houston (DFC2013 and DFC2018)] constituted the HSImix. To facilitate training, we chose the minimum spectral band number 50 of DFC 2018 as the smallest unit for overlapping spectral grouping to generate a huge dataset that has identical spectral bands. The dividing strategy followed [52].

2) Experimental Settings: We compared our CENet with three traditional compression methods, namely, KLT+JPEG2000 [14], 3D-Speck [16], and EMPR [18], and two state-of-the-art deep learning-based methods, namely, Context [23] and Cheng2020att [53]. We implemented all deep learning-based compression methods with the Pytorch framework, in which all models were trained with the Adam optimizer on NVIDIA GTX 3090 GPUs. Moreover, all codecs were running on the same CPU (i9-10920X CPU @3.5 GHz) when inferring. The learning rate was followed [28]. The data augmentation methods involving random cropping, random flip, and random rotating were employed. The training iterations were 60k with three minibatches at a time. The traditional compression methods were applied with open software or codes, where JPEG2000 was implemented with Kakadu version 8.0.5, 3D-Speck was employed by QccPack, and EMPR was conducted by combining with JPEG2000 via the discrete Haar transform.

D. Experimental Results

In this article, all deep learning-based methods were optimized with the PSNR metric. The performance of all compression approaches was evaluated by calculating the average RD performance measured by the PSNR, MS-SSIM, and SAM. All visualization examples, except for CAVE and Bil with a spatial size of 512 × 512, had a spatial size of 256 × 256.

Fig. 5 shows the RD curves of different compression methods on Chikusei using the PSNR, MS-SSIM, and SAM from left to right. Here, CENet denotes the proposed method, and CENet-wo-dualatt is represented as the ablation study in which the CENet is conducted without the interactive dual attention module. Moreover, the baseline of CENet is referred to as mean scale [28], while Context [23] and Cheng2020att [53] are two state-of-the-art deep learning-based methods. As expected, our CENet (plotted with red lines in Figures) is competitive and performs the best compared to most of the existing methods. Even if optimized without the interactive dual attention, the CENet-wo-dualatt still yielded an average PSNR of 29.053 dB at 0.2248 bpppb, exceeding the baseline by 5.42%, that is, the edge features are indeed beneficial for latent representation. When combined with the interactive dual attention module, the CENet outperformed the baseline by 10.99%, revealing that our interactive dual attention module could provide the CompNet with such an ability of rate allocation that benefit for compression performance. In addition,

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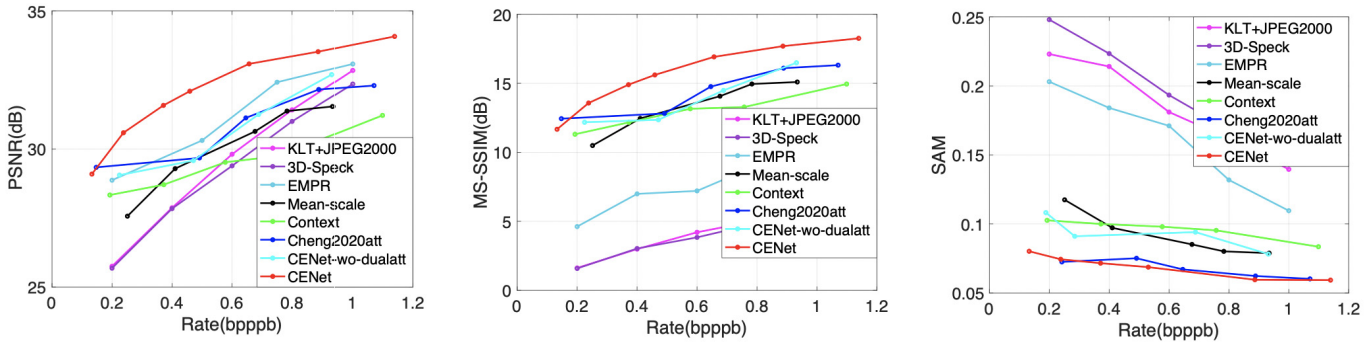


Fig. 5. RD curves in terms of the PSNR, MS-SSIM, and SAM aggregated over the Chikusei dataset. The left plot shows PSNR. The middle plot depicts the MS-SSIM. The right plot presents the SAM. Our proposed CENet (red line) yields a better RD performance compared to all existing approaches.

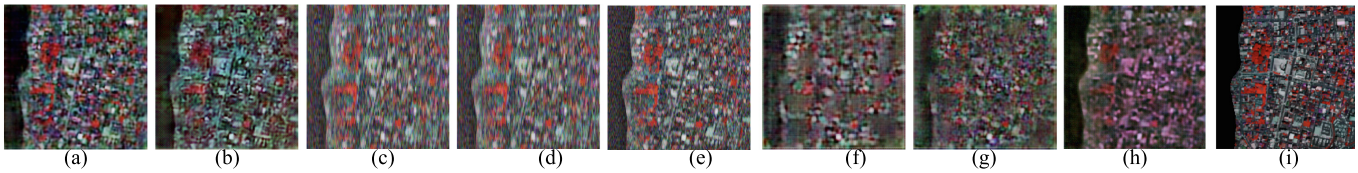


Fig. 6. Visual comparisons over an example patch of the Chikusei dataset at low bit rates. The false color is made with bands (49, 5, and 20). (a) CENet (0.1846 bpppb; PSNR: 26.99 dB; MS-SSIM: 0.9880; and SAM: 0.1955. (b) CENet-wo-dualatt (0.2172 bpppb; PSNR: 24.27 dB; MS-SSIM: 0.9774; and SAM: 0.2963. (c) KLT+JPEG2000 (0.2 bpppb; PSNR: 25.75 dB; MS-SSIM: 0.3110; and SAM: 0.2231. (d) 3D-Speck (0.2 bpppb; PSNR: 25.68 dB; MS-SSIM: 0.3054; and SAM: 0.2481. (e) EMPR (0.2 bpppb; PSNR: 26.12 dB; MS-SSIM: 0.6202; and SAM: 0.2141. (f) Mean-scale (0.2191 bpppb; PSNR: 22.28 dB; MS-SSIM: 0.9679; and SAM: 0.2937. (g) Context (0.1933 bpppb; PSNR: 21.96 dB; MS-SSIM: 0.9716; and SAM: 0.3187. (h) Cheng2020att (0.2016 bpppb; PSNR: 24.60 dB; MS-SSIM: 0.9790; and SAM: 0.2648. (i) Original.

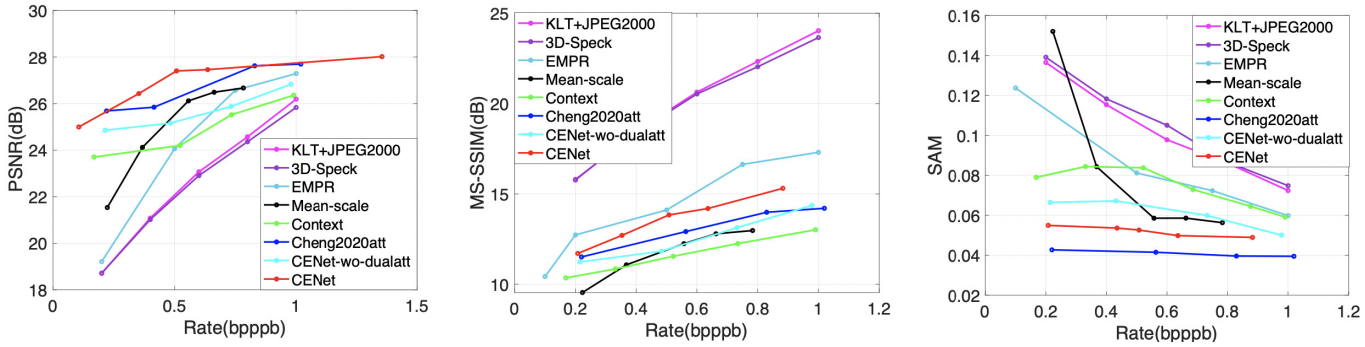


Fig. 7. RD curves in terms of PSNR, MS-SSIM, and SAM aggregated over the Botswana dataset. Our proposed CENet (red line) yields a better PSNR RD performance compared to all existing approaches.

the MS-SSIM and SAM values are still competitive, even if the CENet is not specifically optimized for these two metrics. Fig. 6 presents visual comparisons on one of the test patches of Chikusei that includes numerous tiny buildings at a relatively low bit rate, where quantitative results of PSNR, MS-SSIM, and SAM are also provided below of the corresponding visualization. We chose comparison images produced by other methods with a close bit rate since targeting a precise bit rate is difficult for most compression methods. The results show that both the classical codecs (KLT+JPEG2000, 3D-Speck, and EMPR) and deep learning-based methods suffer from different artifacts. Nonetheless, our CENet clearly displays the best visual perception compared to other compression methods.

We also performed experiments on the other six datasets, involving different object scales and types, to further illustrate the effectiveness of our CENet. Fig. 7 presents the RD curves averaged over the test patches of Botswana dataset and the results demonstrate that our CENet behaves the best when measured with the PSNR metric. In the Botswana dataset, water around reefs and different types of grasses occupy a governing position, leading to a very irregular and detailed structure. In this situation, our CENet can have a compression performance gain with only a small amount of edge information by the interactive dual attention module to offset the structure distortion during compression. In terms of the RD curves using MS-SSIM, CENet did not present the optimal performance as the RD curves with the PSNR. Meanwhile, the visual comparisons in Fig. 8 verified that the visual perception of traditional codecs seemed to be better than that of deep learning-based methods. There are two probable reasons for the above issues. First, this type of data is insufficient during training as all deep learning-based methods suffer from similar artifacts,

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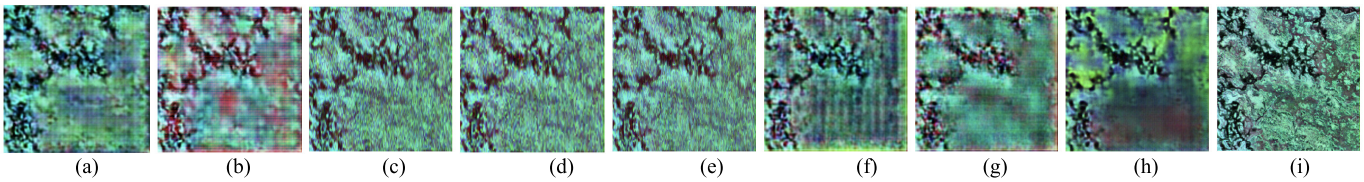


Fig. 8. Visual comparisons over an example patch of the Botswana dataset at low bit rates. The false color is made with bands (49, 5, and 20). (a) CENet (0.1853 bpppb; PSNR: 22.64 dB; MS-SSIM: 0.9325; and SAM: 0.0549. (b) CENet-wo-dualatt (0.2012 bpppb; PSNR: 17.85 dB; MS-SSIM: 0.9247; and SAM: 0.0664. (c) KLT+JPEG2000 (0.2 bpppb; PSNR: 18.70 dB; MS-SSIM: 0.9735; and SAM: 0.1365. (d) 3D-Speck (0.2 bpppb; PSNR: 18.71 dB; MS-SSIM: 0.9737; and SAM: 0.1391. (e) EMPR (0.2 bpppb; PSNR: 21.06 dB; MS-SSIM: 0.9857; and SAM: 0.1153. (f) Mean-scale (0.2030 bpppb; PSNR: 17.53 dB; MS-SSIM: 0.8889; and SAM: 0.1520. (g) Context (0.1918 bpppb; PSNR: 18.04 dB; MS-SSIM: 0.9078; and SAM: 0.0790. (h) Cheng2020att (0.2198 bpppb; PSNR: 19.68 dB; MS-SSIM: 0.9294; and SAM: 0.0427. (i) Original.

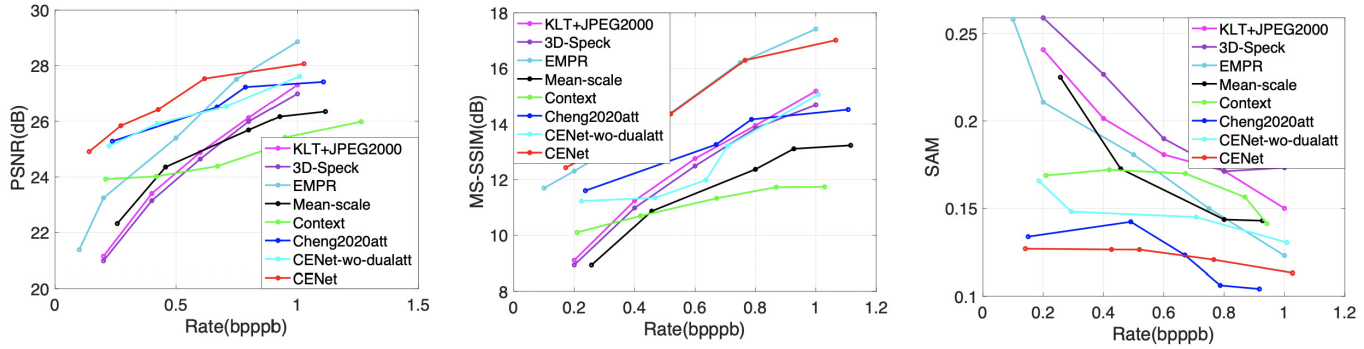


Fig. 9. RD curves in terms of PSNR, MS-SSIM, and SAM aggregated over the Washington DC dataset. Our proposed CENet (red line) yields a better RD performance compared to all existing approaches.

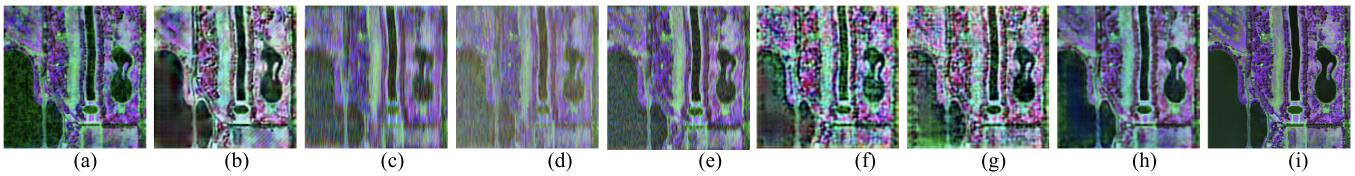


Fig. 10. Visual comparisons over an example patch of the Washington DC dataset at low bit rates. The false color is made with bands (49, 5, and 20). (a) CENet (0.1900 bpppb; PSNR: 23.82 dB; MS-SSIM: 0.9828; and SAM: 0.1707. (b) CENet-wo-dualatt (0.2104 bpppb; PSNR: 18.70 dB; MS-SSIM: 0.9463; and SAM: 0.3158. (c) KLT+JPEG2000 (0.2 bpppb; PSNR: 21.15 dB; MS-SSIM: 0.8770; and SAM: 0.2407. (d) 3D-Speck (0.2 bpppb; PSNR: 20.98 dB; MS-SSIM: 0.8721; and SAM: 0.2590. (e) EMPR (0.2 bpppb; PSNR: 23.40 dB; MS-SSIM: 0.9248; and SAM: 0.2013. (f) Mean-scale (0.2031 bpppb; PSNR: 18.32 dB; MS-SSIM: 0.9368; and SAM: 0.3209. (g) Context (0.1968 bpppb; PSNR: 17.18 dB; MS-SSIM: 0.9317; and SAM: 0.3392. (h) Cheng2020att (0.2098 bpppb; PSNR: 21.32 dB; MS-SSIM: 0.9720; and SAM: 0.1712. (i) Original.

but traditional codecs (i.e., KLT+JPEG2000, 3D-speck, and EMPR) preserve more details since they optimize algorithms with image-specific context rather than data-driven. Second, compared to other deep learning-based methods, our baseline is not strong enough due to its simple architecture. This also reveals two points. First, combined with a simple baseline, the proposed modules can still achieve a competitive performance compared to the previous deep learning-based methods. Second, a strong baseline may result in a better compression performance when incorporated the interactive dual attention and edge-guided loss, which will be illustrated in the following.

In addition, Figs. 9, 11, and 13 show the RD curves averaged over the test patches of Washington DC, Pavia, and Houston, respectively, and Figs. 10, 12, and 14 corresponds to their visual comparisons. These datasets include larger scale buildings or roads, which are convenient for object structure observation. We also chose two datasets with obvious edge structure, namely, CAVE (Figs. 15 and 16), and the unpublished agriculture dataset, namely, Bil (Figs. 17 and 18). Furthermore, to demonstrate the effectiveness of the adaptive edge-guided compression strategy, we exhibited the edges of CAVE obtained from the original HSI with or without interactive dual attention in Fig. 19 to visualize what are good edges when compression performance is good. The corresponding reconstruction results in Fig. 16(a) and (b) illustrated that the proposed CENet only preserved edge details that are useful for compression performance. Also, the numeric results under the corresponding reconstructions illustrate this point again. Furthermore, different from Fig. 16(a), there are some banding noises in the yellow pepper and the red apple (marked by white squares) in Fig. 16(b), blurring and noises in Fig. 16(c)–(e), and some ringing noises in the surrounding objects in Fig. 16(f) and (g). The middle of the lemon slices in Fig. 16(h) almost disappears compared to the original one [Fig. 16(i)]. Consequently,

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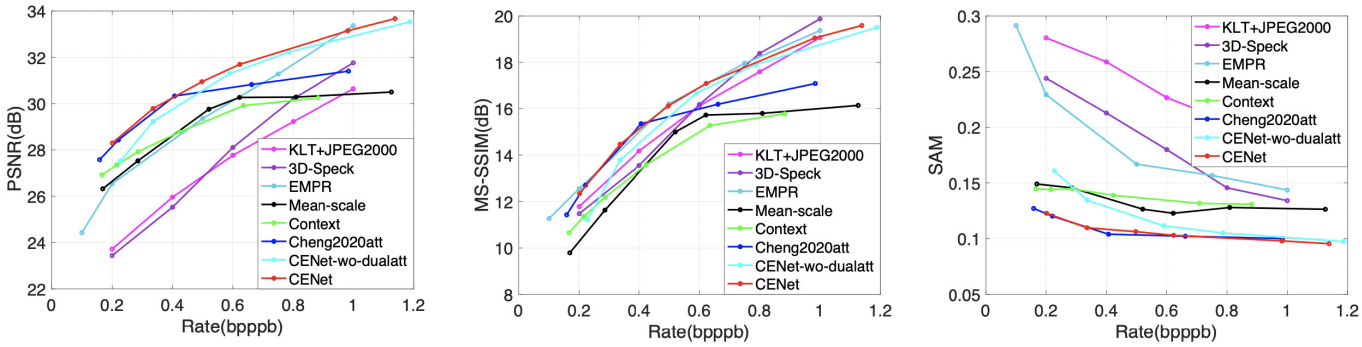


Fig. 11. RD curves in terms of PSNR, MS-SSIM, and SAM aggregated over the Pavia dataset. Our proposed CENet (red line) yields a better RD performance compared to all existing approaches.

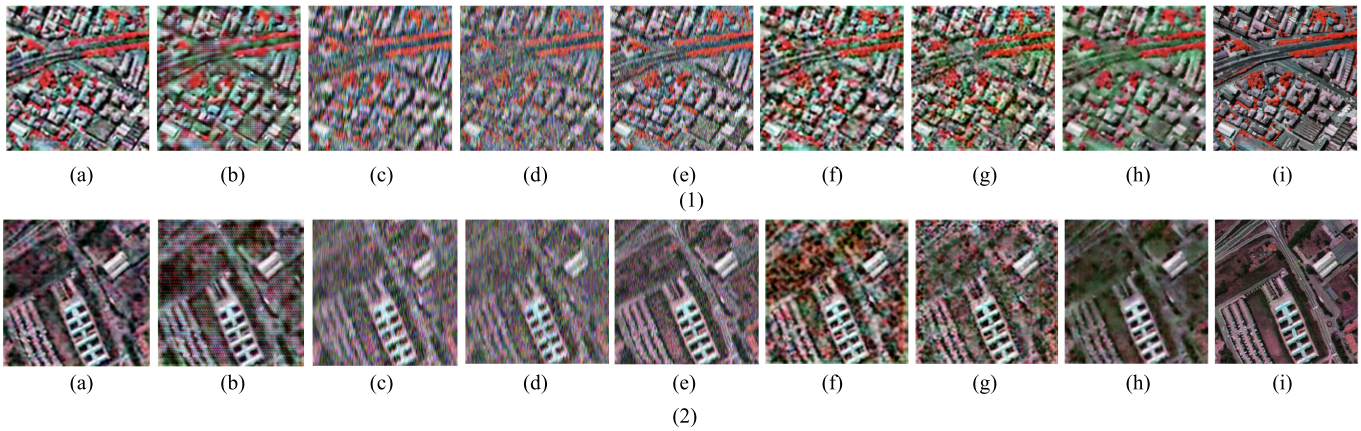


Fig. 12. Visual comparisons over examples of the Pavia dataset at low bit rates. (1) Pavia Centre. (a) CENet (0.1971 bpppb; PSNR: 26.54 dB; MS-SSIM: 0.9897; and SAM: 0.1756. (b) CENet-wo-dualatt (0.2002 bpppb; PSNR: 24.30 dB; MS-SSIM: 0.9463; and SAM: 0.2273. (c) KLT+JPEG2000 (0.2 bpppb; PSNR: 23.72 dB; MS-SSIM: 0.9337; and SAM: 0.2803. (d) 3D-Speck (0.2 bpppb; PSNR: 23.69 dB; MS-SSIM: 0.9336; and SAM: 0.3146. (e) EMPR (0.2 bpppb; PSNR: 25.96 dB; MS-SSIM: 0.9618; and SAM: 0.2587. (f) Mean-scale (0.2010 bpppb; PSNR: 24.07 dB; MS-SSIM: 0.9797; and SAM: 0.2861. (g) Context (0.2099 bpppb; PSNR: 23.48 dB; MS-SSIM: 0.9796; and SAM: 0.3041. (h) Cheng2020att (0.2084 bpppb; PSNR: 23.02 dB; MS-SSIM: 0.9774; and SAM: 0.3671. (i) Original. (2) Pavia University. (a) CENet (0.2036 bpppb; PSNR: 27.93 dB; MS-SSIM: 0.9931; and SAM: 0.1916. (b) CENet-wo-dualatt (0.2080 bpppb; PSNR: 26.07 dB; MS-SSIM: 0.9899; and SAM: 0.3079. (c) KLT+JPEG2000 (0.2 bpppb; PSNR: 24.55 dB; MS-SSIM: 0.9460; and SAM: 0.3293. (d) 3D-Speck (0.2 bpppb; PSNR: 24.36 dB; MS-SSIM: 0.9438; and SAM: 0.3579. (e) EMPR (0.2 bpppb; PSNR: 27.35 dB; MS-SSIM:0.9723; and SAM: 0.2661. (f) Mean-scale (0.2156 bpppb; PSNR: 25.37 dB; MS-SSIM: 0.9853; and SAM: 0.3011. (g) Context (0.2042 bpppb; PSNR: 23.68 dB; MS-SSIM: 0.9832; and SAM: 0.2594. (h) Cheng2020att (0.2120 bpppb; PSNR: 25.69 dB; MS-SSIM: 0.9885; and SAM: 0.3236. (i) Original. The false color is made with bands (49, 5, and 20).

all-round comparisons revealed the advantages of the proposed CENet.

E. Analysis of Various CompNets

As mentioned in Section IV-D, a stronger backbone theoretically produces a better performance. Accordingly, we performed an ablation with Cheng2020att [53] since it was a more advanced version (i.e., improved with residual and attention blocks and discretized Gaussian mixture likelihoods in the entropy model) compared to our mean-scale baseline. We evaluated the compression performance aggregated over two datasets (i.e., Botswana and Houston). Figs. 20 and 21 show an intuitive understanding of the benefits of interactive dual attention and edge-guided loss. The blue lines in the subfigures denote the RD curves obtained by Cheng2020att [53]. The green lines, named Cheng2020att-edge, represent the RD curves obtained by CENet without interactive dual attention. Finally, the red lines depict the RD curves obtained by our proposed CENet (use Cheng2020att as the compression backbone combined with our interactive dual attention and edge-guided loss).

The results in Figs. 20 and 21 show that the performance of Cheng2020att can be significantly improved when combined with the interactive dual attention and edge-guided loss (e.g., yielding an average PSNR of 26.61 dB at 0.2006 bpppb, which distinctly exceeds that of the baseline 25.68 dB at 0.2198 bpppb for Botswana). Specifically, in Fig. 20, the average PSNRs on the Botswana of CENet (red lines) outperformed those of Cheng2020att (blue lines) by 6.94% and Cheng2020att-edge (green lines) by 2.70%. The average of MS-SSIMs increased by 18.26% and 2.77%, while that of SAMs decreased by 4.47% and 1.79%. In Fig. 21, the average PSNRs over the Houston of CENet (red lines) outperformed those of Cheng2020att (blue lines) by 6.09% and Cheng2020att-edge (green lines) by 2.73%. The average of MS-SSIMs increased by 21.23% and 5.52%, while that of SAMs decreased by 18.4% and 15.68%. The experiment

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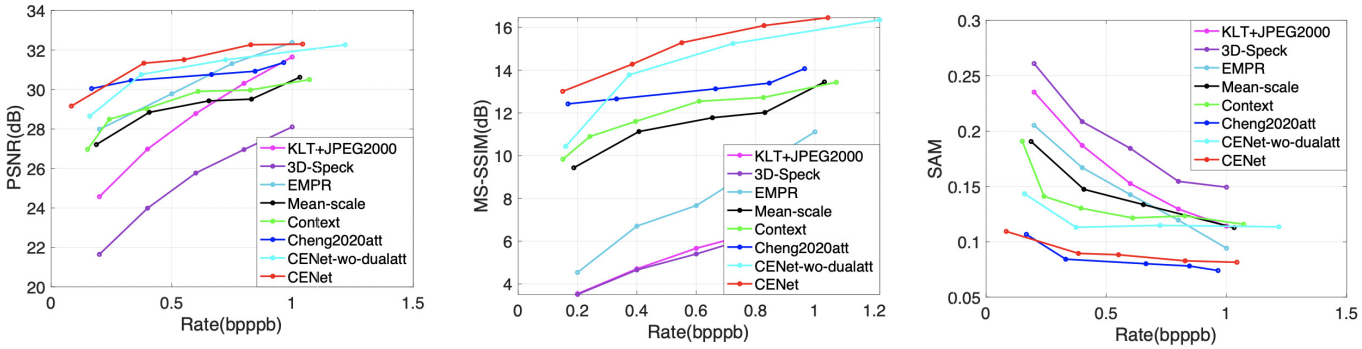


Fig. 13. RD curves in terms of PSNR, MS-SSIM, and SAM aggregated over the Houston dataset. Our proposed CENet (red line) yields a better PSNR and MS-SSIM RD performance compared to all existing approaches.

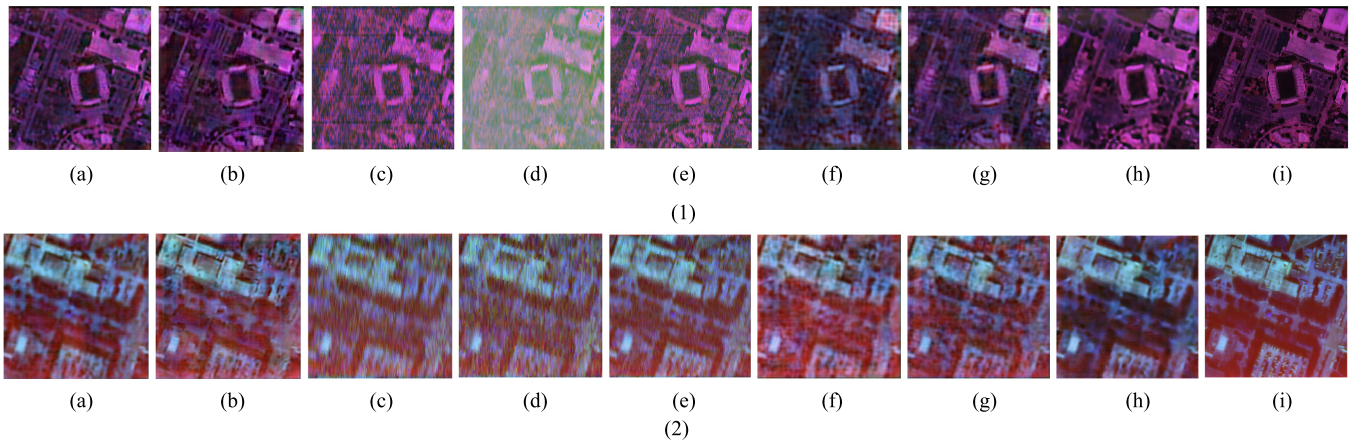


Fig. 14. Visual comparisons over examples of the Houston dataset at low bit rates. (1) DFC2013. (a) CENet (0.1906 bpppb; PSNR: 28.73 dB; MS-SSIM: 0.9702; and SAM: 0.0782. (b) CENet-wo-dualatt (0.2033 bpppb; PSNR: 26.78 dB; MS-SSIM: 0.9547; and SAM: 0.1136. (c) KLT+JPEG2000 (0.2 bpppb; PSNR: 24.57 dB; MS-SSIM: 0.2243; and SAM: 0.2352. (d) 3D-Speck (0.2 bpppb; PSNR: 20.43 dB; MS-SSIM: 0.2256; and SAM: 0.2610. (e) EMPR (0.2 bpppb; PSNR: 26.98 dB; MS-SSIM: 0.4914; and SAM: 0.1870. (f) Mean-scale (0.2084 bpppb; PSNR: 23.47 dB; MS-SSIM: 0.9082; and SAM: 0.1821. (g) Context (0.2043 bpppb; PSNR: 25.55 dB; MS-SSIM: 0.9392; and SAM: 0.1252. (h) Cheng2020att (0.1928 bpppb; PSNR: 28.23 dB; MS-SSIM: 0.9623; and SAM: 0.0812. (i) Original. (2) DFC2018. (a) CENet (0.1953 bpppb; PSNR: 25.98 dB; MS-SSIM: 0.9832; and SAM: 0.1034. (b) CENet-wo-dualatt (0.1960 bpppb; PSNR: 24.63 dB; MS-SSIM: 0.9745; and SAM: 0.1363. (c) KLT+JPEG2000 (0.2 bpppb; PSNR: 22.94 dB; MS-SSIM: 0.8896; and SAM: 0.1799. (d) 3D-Speck (0.2 bpppb; PSNR: 22.86 dB; MS-SSIM: 0.8825; and SAM: 0.1951. (e) EMPR (0.2 bpppb; PSNR: 25.20 dB; MS-SSIM: 0.9425; and SAM: 0.1266. (f) Mean-scale (0.1958 bpppb; PSNR: 22.43 dB; MS-SSIM: 0.9695; and SAM: 0.1370. (g) Context (0.1923 bpppb; PSNR: 22.48 dB; MS-SSIM: 0.9713; and SAM: 0.1443. (h) Cheng2020att (0.2061 bpppb; PSNR: 24.67 dB; MS-SSIM: 0.9792; and SAM: 0.1671. (i) Original. The false color is made with bands (49, 5, and 20).

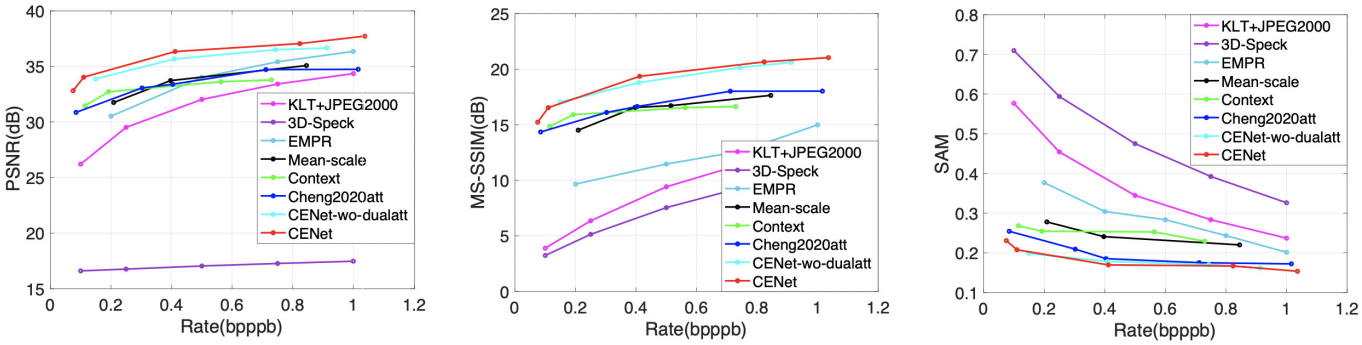


Fig. 15. RD curves in terms of PSNR, MS-SSIM, and SAM aggregated over the CAVE dataset. Our proposed CENet (red line) yields a better RD performance compared to all existing approaches.

also indicates that our interactive dual attention module and edge-guided loss support plug and play, which can be easily merged with existing deep learning-based compression architectures to further boost performance.

F. Model Complexity

The training time and the inference time that in the encoding and decoding phases are shown in Table II. Moreover, the memory cost (measured with a number of parameters), the

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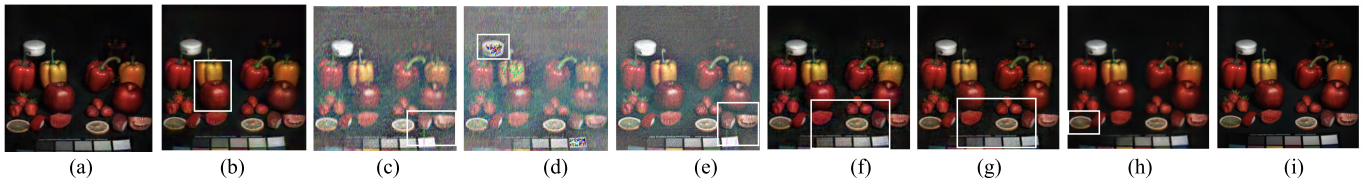


Fig. 16. Visual comparisons over an example of the CAVE dataset at low bit rates. The false color is made with bands (15, 5, and 30). The white squares are used for marking artifacts or smoothing. (a) CENet (0.2112 bpppb; PSNR: 37.21 dB; MS-SSIM: 0.9865; and SAM: 0.1449. (b) CENet-wo-dualatt (0.2206 bpppb; PSNR: 33.00 dB; MS-SSIM: 0.9667; and SAM: 0.2393. (c) KLT+JPEG2000 (0.25 bpppb; PSNR: 30.06 dB; MS-SSIM: 0.7468; and SAM: 0.4475. (d) 3D-Speck (0.25 bpppb; PSNR: 19.76 dB; MS-SSIM: 0.6644; and SAM: 0.5731. (e) EMPR (0.25 bpppb; PSNR: 32.03 dB; MS-SSIM: 0.8855; and SAM: 0.3446. (f) Mean-scale (0.2227 bpppb; PSNR: 32.44 dB; MS-SSIM: 0.9614; and SAM: 0.2476. (g) Context (0.2318 bpppb; PSNR: 34.15 dB; MS-SSIM: 0.9744; and SAM: 0.2165. (h) Cheng2020att (0.2437 bpppb; PSNR: 33.31 dB; MS-SSIM: 0.9666; and SAM: 0.1849. (i) Original.

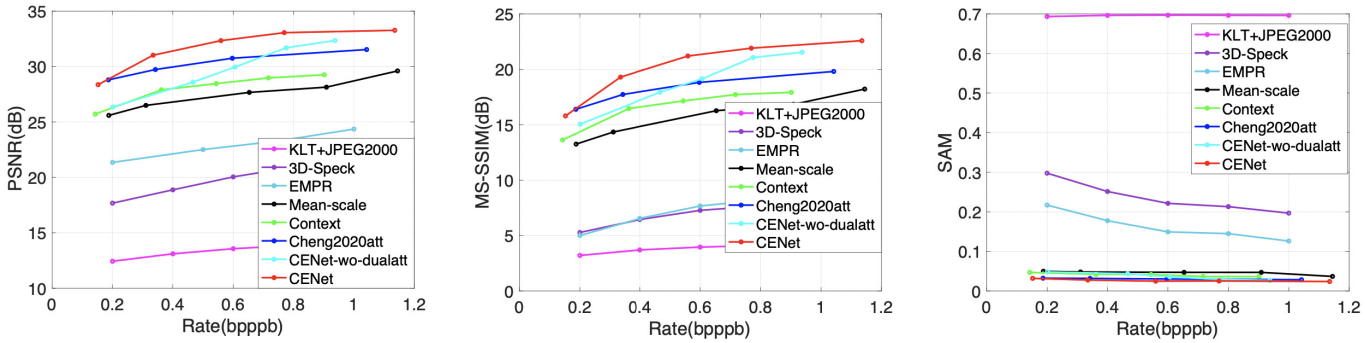


Fig. 17. RD curves in terms of PSNR, MS-SSIM, and SAM aggregated over the Bil dataset. Our proposed CENet (red line) yields a better RD performance compared to all existing approaches.

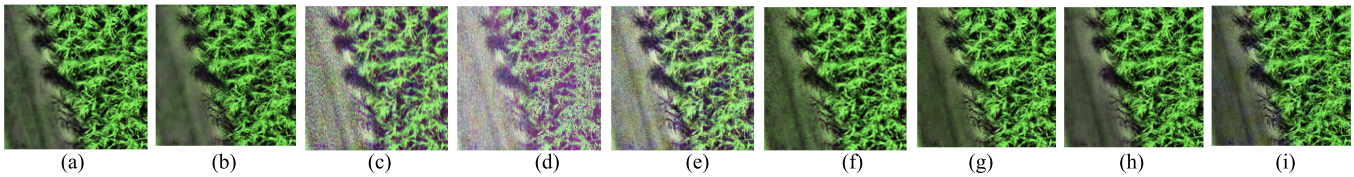


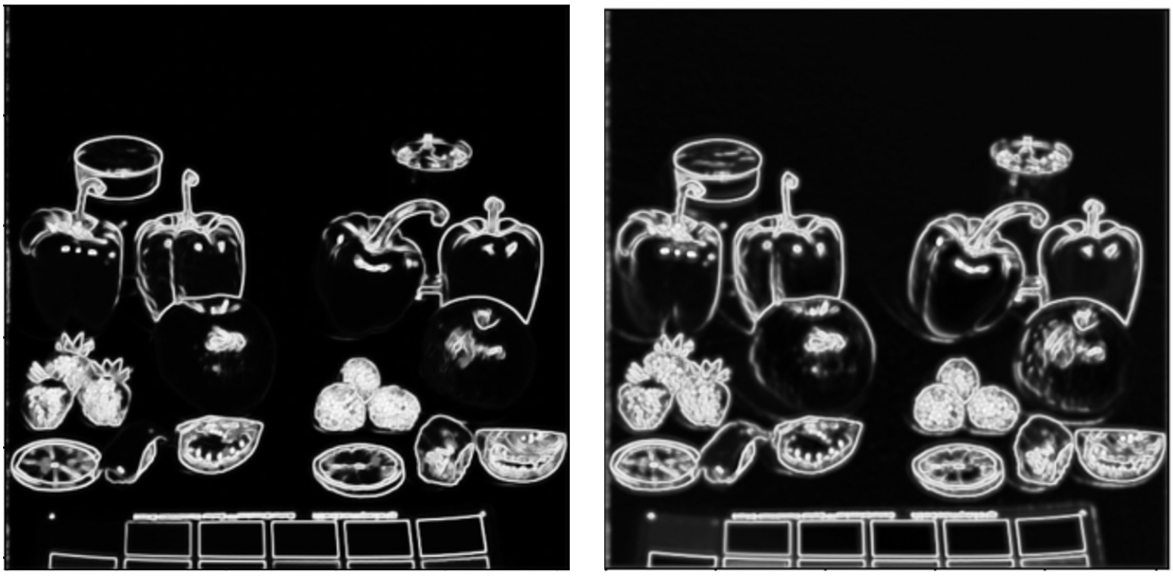
Fig. 18. Visual comparisons over an example patch of the Bil dataset at low bit rates. The false color is made with bands (22, 86, and 2). (a) CENet (0.1518 bpppb; PSNR: 28.76 dB; MS-SSIM: 0.9697; and SAM: 0.0412. (b) CENet-wo-dualatt (0.1736 bpppb; PSNR: 27.44 dB; MS-SSIM: 0.9592; and SAM: 0.0559. (c) KLT+JPEG2000 (0.2 bpppb; PSNR: 24.42 dB; MS-SSIM: 0.7535; and SAM: 0.1441. (d) 3D-Speck (0.2 bpppb; PSNR: 17.68 dB; MS-SSIM: 0.7036; and SAM: 0.2971. (e) EMPR (0.2 bpppb; PSNR: 26.81 dB; MS-SSIM:0.8430; and SAM: 0.1231. (f) Mean-scale (0.1672 bpppb; PSNR: 25.36 dB; MS-SSIM: 0.9436; and SAM: 0.0727. (g) Context (0.1924 bpppb; PSNR: 26.33 dB; MS-SSIM: 0.9630; and SAM: 0.0687. (h) Cheng2020att (0.1712 bpppb; PSNR: 28.49 dB; MS-SSIM: 0.9654; and SAM: 0.0436. (i) Original.

running speed (measured with fps), and the computational cost (measured with FLOPs) for all methods are also provided in Table II to show the complexity of each method, intuitively. The symbol "–" denotes that there is no result because the parameters for traditional codecs are given by handcrafting rather than learning. From Table II, we can find that the inference time of the proposed CENet that in the encoding and decoding phases is competitive when processing HSIs with different spectral bands or spatial size even if our code has not been particularly optimized. Moreover, the running speed (fps measured by GPUs) illustrates this point again. Note that increasing encoding bit rate does not obviously increase the inference time for deep learning-based methods and KLT+JPEG2000, but that of 3D-speck and EMPR prolong as bit rate increases. Therefore, we only presented the relatively optimal running time of 3D-speck in Table II. In addition, there is no surprising that the parameters, the training time, and the FLOPs of the proposed CENet are larger than others since there is a need for edge extraction and a measure of how much structure detail is needed for each compressed latent by interactive dual attention. We did not strive to save the capacity of the CENet to reduce computational complexity as we aim to determine the potential of edge features for compression performance instead of trading off complexity against performance.

G. Discussion

The CENet model, a unified framework in which a compression architecture is embedded with an edge extraction neural network, targets adaptively learning a strong structural latent representation with the help of interactive dual attention as driven by the edge-guided loss. As shown in Figs. 5–18, compared to existing compression methods, the superiority of the CENet is evident with the increasing PSNR, MS-SSIM, and decreasing SAM. The proposed CENet has the ability to distinguish what is useful edge for compression performance. Moreover, the edge-guided loss of CENet can optimize the

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(a) (b)

Fig. 19. Edges of CAVE obtained from original HSI with or without interactive dual attention. (a) Edges with interactive dual attention. (b) Edges without interactive dual attention.

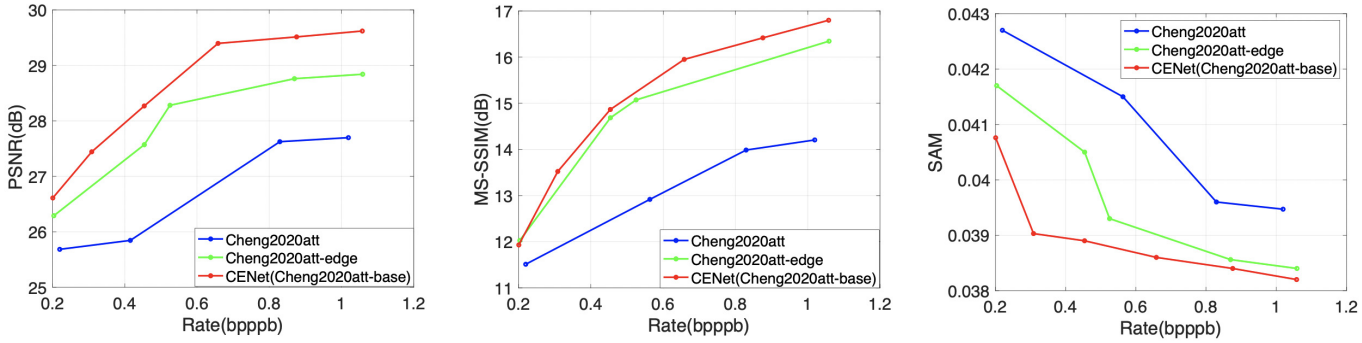


Fig. 20. RD curves in terms of PSNR, MS-SSIM, and SAM aggregated over the Botswana dataset with a stronger compression backbone, Cheng2020att [53]. The proposed interactive dual attention and edge-guided loss modules combined with Cheng2020att show a distinct rise compared to the baseline Cheng2020att in all three metrics.

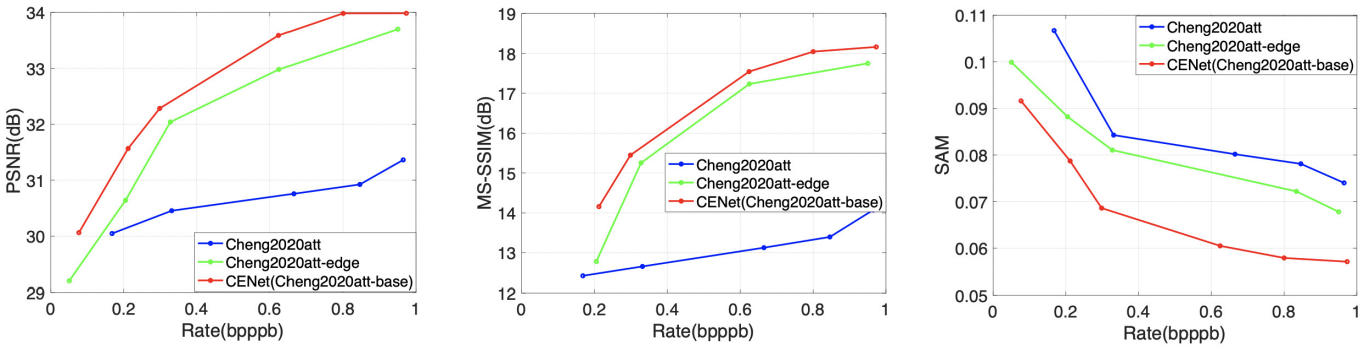


Fig. 21. RD curves in terms of PSNR, MS-SSIM, and SAM aggregated over the Houston dataset with a stronger compression backbone network, Cheng2020att [53]. The proposed interactive dual attention and edge-guided loss modules combined with Cheng2020att show a distinct rise compared to the baseline Cheng2020att in all three metrics.

CompNet and EdgeNet simultaneously. Figs. 20 and 21 show that our CENet can give a distinct improvement, even with a simple compression architecture. In addition, the interactive dual attention and edge-guided loss can also be easily made compatible to different compression architectures to satisfy the practical reconstruction quality demand. The CENet can be regarded as a supervised compression architecture. On the one hand, pixels in reconstruction can be better connected without introducing too much bit rates by selectively delivering structural representation via interactive dual attention. On the other

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TABLE II

TRAINING TIME, THE ENCODING AND DECODING TIME DURING INFERRING, AVERAGE RUNNING SPEED (FPS), THE COMPUTATIONAL COST (FLOPS), AND THE MEMORY COST (THE NUMBER OF PARAMETERS) ON THREE GROUPS OF HSIS FOR THE PROPOSED CENET AND OTHER COMPARISON METHODS

Datasets	Methods	Parameters(M)	FPS	FLOPs(G)	Training time(h)	Enc-time(s)	Dec-time(s)
CAVE	KLT+JPEG2000	–	–	–	–	2.68	1.36
	3D-Speck(0.25bpppb)	–	–	–	–	6.19	5.84
	EMPR (0.25bpppb)	–	–	–	–	61.23	69.15
	Mean-scale(base)	17.60	13	155	0.5	0.55	0.73
	Context	20.42	6	158	0.9	5.15	14.37
	Cheng2020att	27.80	6	278	1.5	5.94	16.96
	CENet-wo-dualatt	33.51	7.5	420	3.4	0.50	0.80
	CENet	35.20	6	572	3.8	2.70	0.65
Bil	KLT+JPEG2000	–	–	–	–	3.45	2.70
	3D-Speck(0.25bpppb)	–	–	–	–	25.31	19.85
	EMPR(0.25bpppb)	–	–	–	–	64.57	70.42
	Mean-scale(base)	18.51	10	301	0.7	0.82	1.45
	Context	21.33	10	304	1.2	5.37	15.15
	Cheng2020att	28.64	5	328	2.6	6.17	17.28
	CENet-wo-dualatt	34.54	7	440	4.0	0.83	1.51
	CENet	36.22	6	596	4.5	3.35	1.43
HSImix	KLT+JPEG2000	–	–	–	–	1.12	0.87
	3D-Speck(0.25bpppb)	–	–	–	–	3.32	4.15
	EMPR(0.25bpppb)	–	–	–	–	32.41	29.76
	Mean-scale(base)	17.78	6	47	0.6	0.12	0.15
	Context	20.60	6	48	1.0	0.54	1.69
	Cheng2020att	27.97	5	73	2.3	0.63	1.47
	CENet-wo-dualatt	33.72	7.5	106	3.6	0.10	0.11
	CENet	35.40	6	144	4.2	0.45	0.13

hand, the compression model is optimized with edge-guided loss to guarantee model learning in a more structural direction.

V. CONCLUSION

This study proposed an edge-guided HSI CompNet, called CENet, for high-quality reconstruction. Specifically, we embedded an EdgeNet into a CompNet architecture to guide compression optimization by the edge-guided loss, thereby improving the structural ability of the latent representation. Furthermore, we proposed interactive dual attention to help the EdgeNet adaptively capture rich context with strong continuous feature relations since some edges had little contribution for reducing distortion but increasing bit rates. The quantitative and visual analysis results on seven different HSI datasets illustrated that the introduced interactive dual attention and edge-guided loss could adaptively preserve edges that can promote compression performance. More importantly, the proposed interactive dual attention and edge-guided loss could be inserted in any compression neural network for performance boost. However, the CENet only considers edge features, which may be insufficient for addressing some edge-inadequate HSIs. In our future work, we will attempt to combine more task-level features into compression optimization to narrow the information gap between the latent representation and the task-specific requirements.

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